PRO-3015 · 6 & 12 KW · COMPACT FIBER LASER TECHNOLOGY

GFL FIBER LASER TECHNICAL SPECIFICATION

YOUR PARTNER IN LASER CUTTING MACHINES



Since 1974
MANUFACTURING EXPERIENCE

6 & 12 kW
RAYCUS · FSCUT

1.5 G
X-Y ACCELERATION

Canada
PARTS & TECHNICAL SUPPORT

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FULL TECHNICAL SPECIFICATION · DENER PRO 3015 · 6 & 12 KW

CONTENTS

1 DENER Machinery 03
2 Machine information 04
2.1 Machine concept 04
3 Fiber laser advantages 05
4 Design and manufacturing advantages 06
5 Machine components and options 07
5.1 Control unit 20
5.1.1 FSCUT 20
5.1.2 FANUC (option) 23
5.2 Power sources (resonator) 10
5.2.1 Raycus 10
5.2.2 IPG (option) 11
5.3 Cutting head 08
5.3.1 Boji 08
5.3.2 Precitec (option) 09
5.4 Chiller 12
5.5 Rack and pinion system 07
5.6 High-speed Z, X and Y axes 08
5.7 Linear guides 16
5.8 Electric panel components 17
5.9 Pneumatic system 19
5.10 Air conditioner (option) 17
5.11 Nozzle cleaning and calibration 16
5.12 Burr-free cutting parameters 25
5.13 Aspiration system 13
5.14 Shuttle table 14
5.15 Safety system 14
5.16 Scrap boxes 13
5.17 Automatic central lubrication 16
5.18 Front door and pallet door 15
6 Software 22
6.1 CypNest 22
6.2 Metalix 24
7 Transport dimensions 31
Gas control · servo motors · starting kit 18
Other technical specification 26
Electric consumption 27
Configuration 28
Warranty conditions 29
FAQ 30
Installation requirement 31

INCLUDED WITH EVERY NOVOTECH MACHINE

- Made in Turkey with European components
- CSA-approved electrical panel
- 1-year warranty
- **7-day money-back guarantee** (in-stock machines only)
- Spare parts & technical support readily available in Canada
- **Order Today with Just 10% Down — 0% Down Financing Available**
- **Installation and training available**

SOLD & SERVICED IN CANADA
by NovoTech Machine Tools — Mississauga, ON

1 · DENER MACHINERY

ABOUT DENER



Dener campus, Kayseri: machine tool assembly, sheet metal machine fabrication, assembly and welding halls, tooling workshop, machine covers fabrication, headquarters and showroom

Dener Makina, established in 1974 in Kayseri / Turkey, is globally recognized as one of the best sheet metal machine manufacturers. With 800 expert employees and a production capacity of 3,400 machines, Dener constantly improves its production technology.

The company is located on a 14,000,000 sq ft (1,300,000 m^2) site with 1,670,000 sq ft (155,000 m^2) of closed area and exports to more than 100 countries, mainly to European Union countries, the USA and Asia. All Dener machines meet EU standards and a strict quality control is made at every stage of production.

DENER GROUP PRODUCT RANGE — SHEET METAL DIVISION

- Fiber laser cutting machines
- 100% electrical press brakes (widest range)
- Hydraulic press brakes
- Tandem press brakes
- Plasma cutting machines
- Hydraulic shearing machines

MACHINING DIVISION

- Vertical milling machines
- Lathes
- Surface grinding machines



TÜRKIYE

- Dener Makina headquarter – Kayseri
- Representative – Bursa
- Representative – İstanbul
- Representative – İzmir
- Representative – Anatolian region



INDIA

- Dener Makina office & showroom – Pune



USA

- Dener Makina office & showroom – Chicago

2 · MACHINE INFORMATION · 2.1 MACHINE CONCEPT

MACHINE SPECIFICATION

Through optic technology, the DENER fiber laser machine is able to achieve high-speed performance while cutting 2D sheet metal. Equipped with various cutting modes and well-known fiber laser sources, the Dener fiber laser can easily cut challenging parts. A well-thought design allows easy transport and installation, with a machine equipped with user-friendly equipment. Speeds achieved on the X, Y and Z axes are significantly satisfying and allow a fast return on investment combined with cutting precision.



The Dener body frame was designed and manufactured around the user's needs. With a stress-relieved, robust body structure and a bridge designed with space geometry, the machine is ready for high acceleration.

Six pneumatic aspiration compartments are also available on this model to facilitate the suction of the polluted fume resulting from cutting. Small parts that fall during the cutting process can be found in one of the six scrap boxes mounted on wheels that are easy to remove. The machine is adapted to work with an aspiration system that can collect all the fume resulting from the cutting process. The cutting tables were designed to reduce the exchange time, therefore increasing productivity.

3 · FIBER LASER ADVANTAGES

FIBER LASER TECHNOLOGY ADVANTAGES

Compared to CO2 laser technology, fiber laser technology requires less maintenance and its operation consumes less energy. While a CO2 laser requires 3 – 4 hours of maintenance per week, a fiber laser requires far less maintenance.

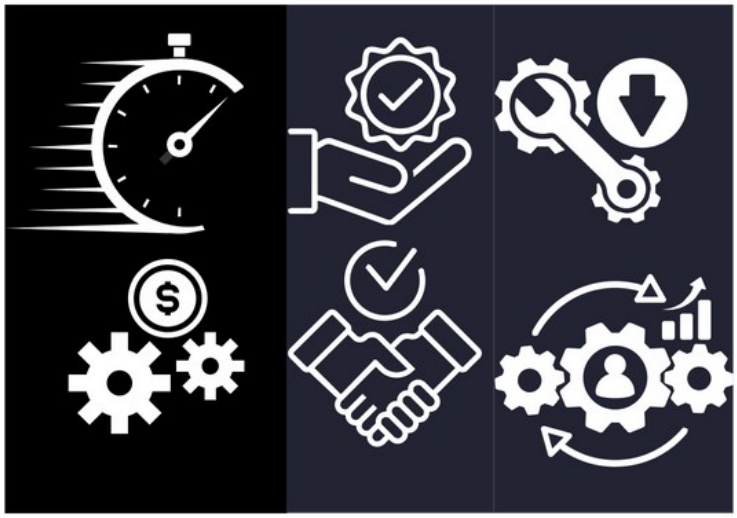
Fiber lasers cut reflective materials such as copper, brass, aluminum and stainless steel, operations that are not possible on a CO2 laser.

Cutting thickness limits on fiber lasers are disappearing over time. A 20 kW laser source is able to cut 2-3/8" (60 mm) mild steel, and higher power sources are also available.

Fiber laser technology is the best alternative to the CO2 laser, as the shorter wavelength and higher power at the focal point make cutting easier. The fiber laser combines a fast return on investment, lower production cost, high profitability and lower maintenance. Contrary to CO2, a fiber laser does not require a warm-up stage before it can cut.

KEYWORDS

- Fast
- Reliable
- High quality cutting
- Affordable
- High technology
- Rigid body frame
- Quality components
- Low operating cost
- Low maintenance cost





Parts cut on a Dener fiber laser

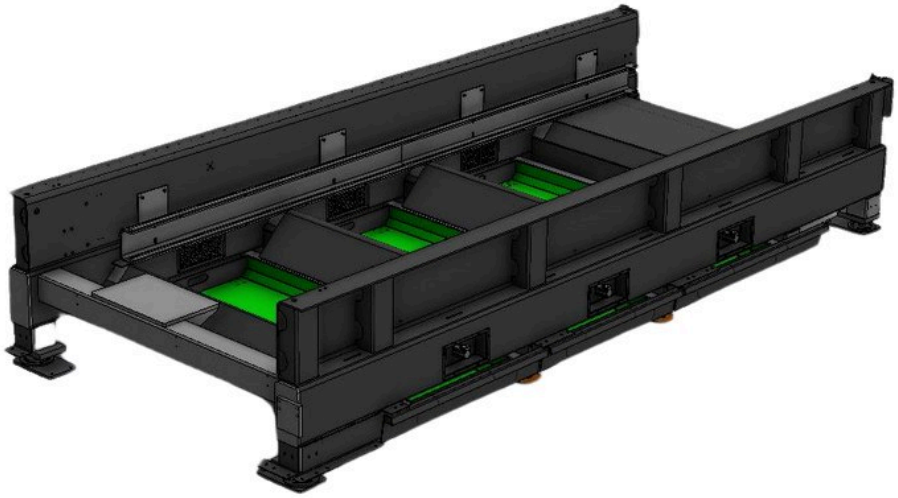
4 · DESIGN AND MANUFACTURING ADVANTAGES

DENER MANUFACTURING

All DENER machines are manufactured with respect to European safety norms. A meticulous control is made at every step of the production line, enhancing the quality of the final product. The ERP system we use gives real-time visibility on the production process. DENER only uses quality components from well-known suppliers.

COMPACT WELDED FRAME

From choosing high-quality steel, followed by the art of welding, the frame passes through a stress-relief process. The remaining machining of the parts is made on a very precise 5-axis portal milling and boring machine. The final frame is rigid, ergonomic and highly resistant to high-speed displacements of the bridge.



BRIDGE

The DENER fiber laser bridge is designed with space geometry and ensures stability and precision during the cutting process. The bridge can easily operate at high speed.



5 · MACHINE COMPONENTS AND OPTIONS

RACK AND PINION · Z AXIS

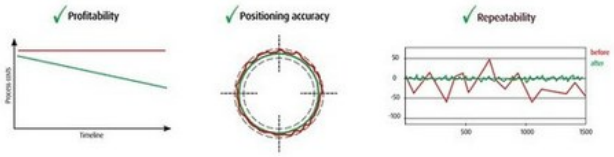
Dener reserves the right to replace components with newer, better-performing systems.

RACK AND PINION SYSTEM FRAMO MORAT

The advanced combination of planetary gearbox, pinion, rack and lubrication systems results in a high-end linear motion system. Positioning accuracy is enhanced, assembly is easy, wear is low and service life is long.

Thanks to these genuine components, displacement on the various axes is accomplished with a positioning tolerance of ±0.002" (±0.05 mm) — positioning of the axis, not cutting precision.





PERFORMANT Z AXIS

The Dener fiber laser Z axis is designed and configured to reach 1 G. The Z axis can handle high acceleration while keeping its cutting accuracy. Ahead of its competitors, the Z axis on the GFL model has a stroke of 13.8" (350 mm).



13.8" (350 MM) STROKE

GFL 3015 L · GFL 4020 L · GFL 6020 L

TECHNICAL INFORMATION

AXIS	GFL 3015 L · 4020 L · 6020 L
X axis	394 ft/min (120 m/min)
Y axis	394 ft/min (120 m/min)
Positioning tolerance	±0.002" (±0.05 mm)
Repeatability	±0.0012" (±0.03 mm)

Note: the information given is for the axes, not for cutting.

5.3 · AUTOMATIC LASER CUTTING HEAD BOJI BLT 421 · STANDARD

The BLT 4 series delivers great performance with stable cutting and simple installation and setup, and is available for power levels below 30 kW. Thanks to the capacitive sensor, the offset can be set directly from the sheet position.



- Brand new optical solutions come with closed-loop auto focusing
- Slag-free cutting, nozzle cooling and water cooling sensor supported — makes cutting more stable and efficient
- **Ready-to-use optics drawer: lens changed in 2 minutes**
- **Groups of internal sensors for real-time closed-loop monitoring**

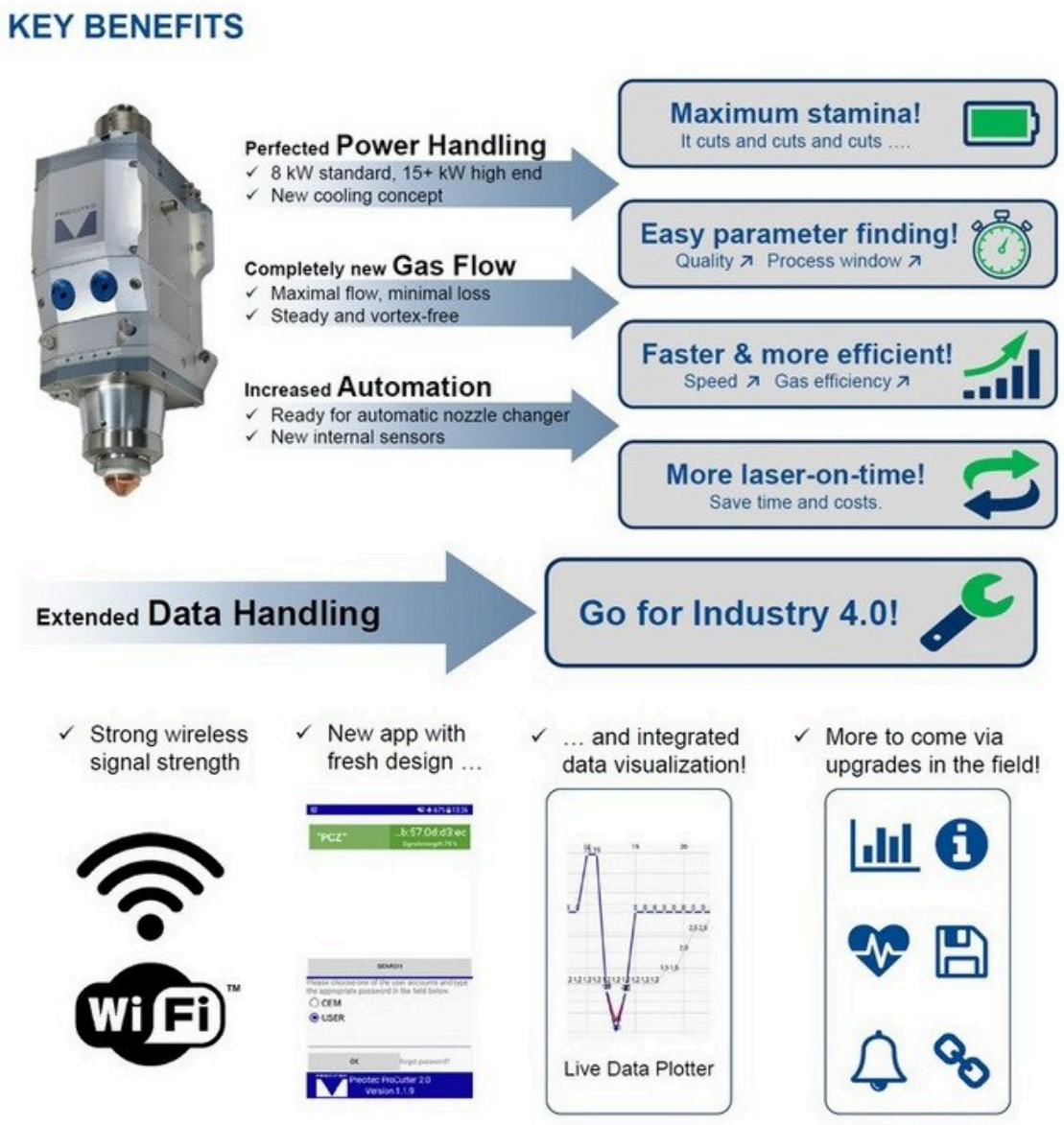
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5.3.2 · CUTTING HEAD · OPTION

PRECITEC PROCUTTER 2.0

With the ProCutter 2.0, cutting processes that were previously hidden away can now be used in an industrially stable manner. It is no longer necessary to choose between high quality and high speed: both are possible – increased machine utilization and a shorter payback period. The amount of rework is reduced to a minimum.



- ✓ Maximum productivity thanks to superior technology
- ✓ Around-the-clock consistent cut quality at high laser power
- ✓ Intelligent sensors at their best

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5.2.1 · POWER SOURCE · STANDARD

RAYCUS FIBER LASER POWER SOURCE

The single-module CW fiber laser is completely researched, developed and produced by Raycus independently. Raycus has an efficient and professional R&D and production team. The single-module CW fiber laser series not only meets the diversified demands of customers to the greatest extent, but also provides a personal customization service. The laser output connector is QBH, with multiple control modes. Its good compatibility has been widely recognized by the market, and the product sells well at home and abroad.

It has the advantages of high electro-optical conversion efficiency, good beam quality, high energy density, wide modulation frequency, strong reliability, long life and maintenance-free operation. It can be widely used in welding, precision cutting, cladding, surface treatment, 3D printing and other fields. Its optical fiber output makes it easier to integrate with robots into flexible manufacturing equipment to meet the needs of metal processing.





From 2 kW to 20 kW
RAYCUS FIBER LASER POWER SOURCES

5.2.2 · POWER SOURCE · OPTIONAL

IPG FIBER LASER POWER SOURCE

YLS-CUT lasers have been developed specifically for demanding cutting applications. Supplied in the smallest form factor available on the market, these lasers can be easily integrated within cutting machines. The lasers are packaged in a hermetically sealed cabinet containing an internal dryer, enabling the laser to be used in the harshest of production environments.

The YLS-CUT incorporates the latest IPG technical improvements in the design of diode modules, fiber blocks, digital power supplies and digital control electronics, resulting in unparalleled reliability and increased control flexibility, stability and precision. The new control system allows integrated control of both the laser and the process subsystem. The industry-leading wall-plug efficiency of over 40% results in electrical cost savings adding up to many tens of thousands of dollars over the lifetime of a laser. Hot redundancy ensures 100% uptime with no change in power, record reliability and maintenance-free operation.

The CUT series lasers are available up to 20 kW output power with a wide variety of fiber delivery options, starting at a 50 µm core diameter. The YLS-U is the latest super-compact hermetic cabinet packaging in the 1 – 10 kW power range, incorporating all the features and technology advancements of the YLS-CUT series. YLS-U series lasers are also available with the High Peak Power (HPP) option, enabling faster and cleaner piercing, increased quality of fine-feature cutting and denser part nesting.



From 2 kW to 20 kW

IPG YLS FIBER LASER POWER SOURCES (OPTIONAL)

5.4 · CHILLER · STANDARD

S&A CHILLER

With more than 20 years of experience, S&A chillers were designed specially for fiber laser technology. The chiller cools the resonator and the cutting head optics. Inlets and outlets are made of stainless steel to ensure the durability of the product. The water pump and all other equipment were also developed for long-life operation. Chillers are matched to the cutting power and come in different sizes.





- Dual cooling circuit
- Active cooling
- Temperature stability: ±1.8 °F (±1 °C)
- Temperature control range: 41 – 95 °F (5 – 35 °C)
- Refrigerant: R-410a
- Intelligent digital control panel
- Integrated alarm functions
- Back-mounted fill port and easy-to-read water level check
- RS-485 Modbus communication function
- High reliability, energy efficiency and durability
- Available in 380 V



Dual water inlet and water outlet

Stainless steel inlet and outlet to prevent corrosion and leakage.



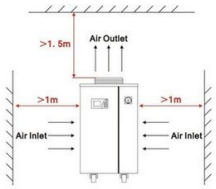
Dual temperature control

The intelligent control panel offers two independent temperature control systems: one controls the temperature of the fiber laser, the other the optics.



Caster wheels for easy mobility

Four caster wheels offer easy mobility and unmatched flexibility.



Ventilation

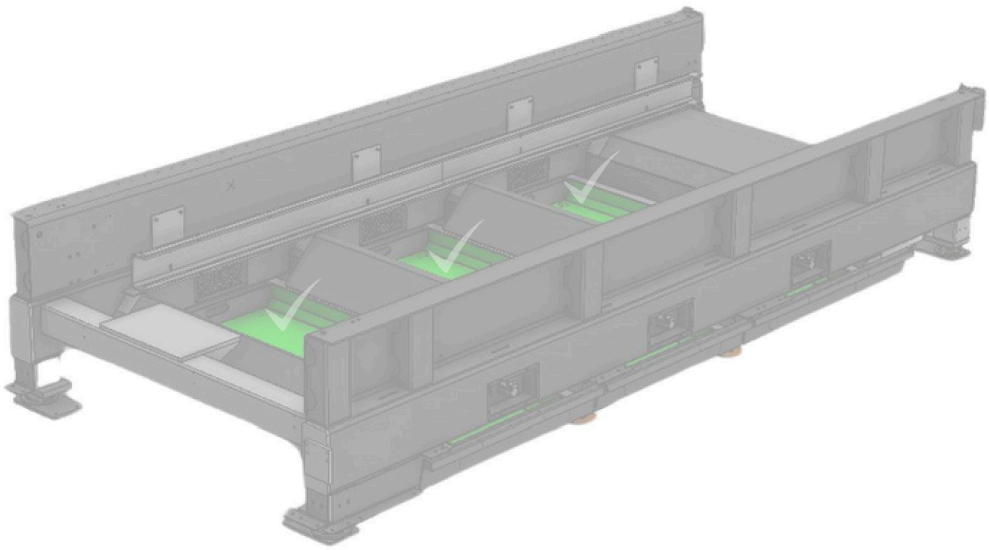
Air inlets and outlets of the chiller: keep the clearances shown round the unit.

5.13 · 5.16 · STANDARD

ASPIRATION CELLS · SCRAP BOXES

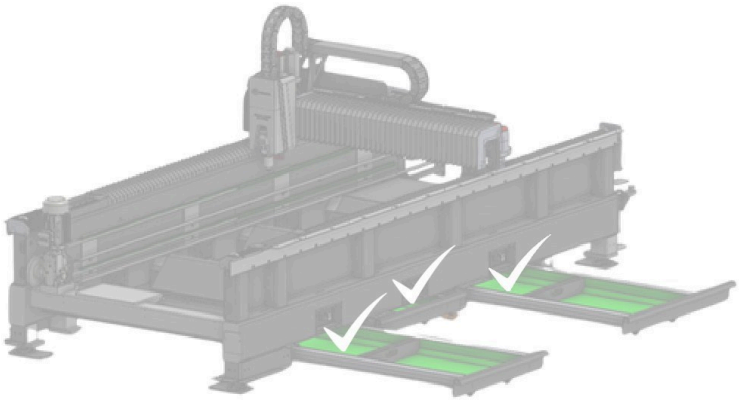
ASPIRATION CELLS STANDARD

DENER fiber laser aspiration was designed to be highly efficient in treating the polluted fume. The aspiration units are activated depending on the cutting position.



SCRAP BOXES STANDARD

With an ergonomic design, the scrap boxes are easy to pull, and small parts and scrap can be easily reached.



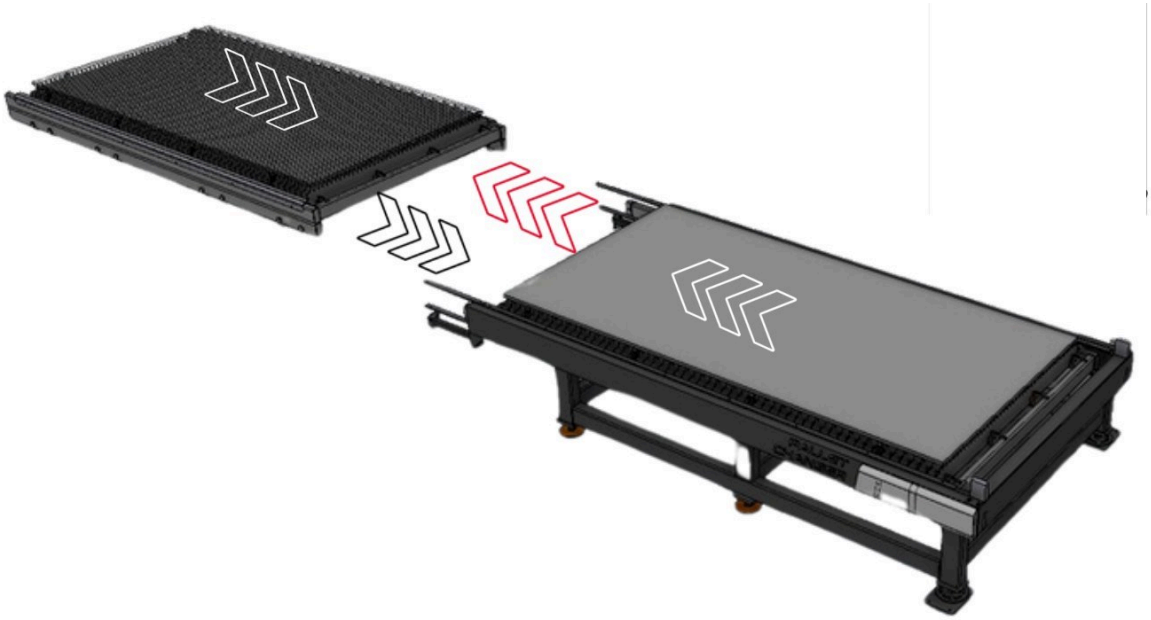
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5.14 · 5.15 · STANDARD

CUTTING TABLES · SAFETY

CUTTING TABLES STANDARD

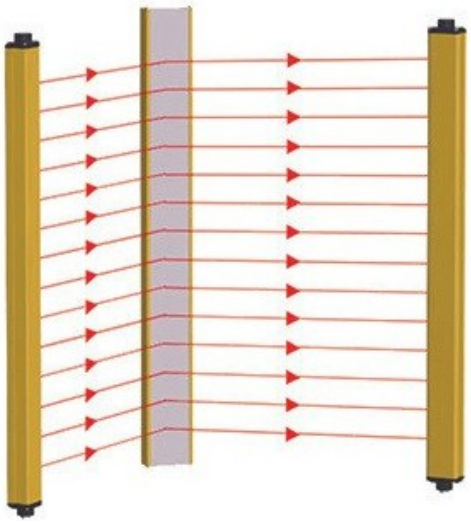
DENER GFL fiber lasers are equipped with a dual exchangeable table. It allows a fast change of table in one motion.



SAFETY · LIGHT CURTAIN STANDARD

Machines systematically stop working when the operator is inside the working region.

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5.15 · 5.18 · STANDARD

EMERGENCY STOPS · DOORS

EMERGENCY STOP BUTTON AND BUZZER STANDARD

An easily accessible emergency stop button provides a safe working environment for the operator. A buzzer on top of the machine shows the working state of the machine in real time.



FRONT SLIDING DOOR AND REAR PNEUMATIC DOOR STANDARD

A front sliding door with special laser glass allows the machine user to operate safely, gives easy access to the cutting head and makes maintenance of the machine easier.

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5.11 · 5.17 · 5.7

NOZZLE CLEANING · CALIBRATION · LUBRICATION · LINEAR GUIDES

NOZZLE CLEANING

With the automatic nozzle cleaning provided, we ensure a better cutting quality and a longer life of the nozzle.

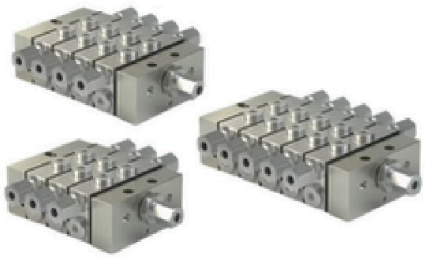
CALIBRATION

The DENER fiber laser calibrates directly on the sheet metal, so the cutting process can start faster.

AUTOMATIC LUBRICATION SYSTEM STANDARD

The Dener fiber laser is equipped with a centralized lubrication system that lubricates the axes according to the travel distance. It ensures that the mechanical parts work smoothly.





LINEAR GUIDES STANDARD · THK

THK linear guides resist heavy loads and ensure stability and precision in the motion of the different axes.



THK
The Mark of Linear Motion

5.8 · 5.10

ELECTRICAL COMPONENTS

DENER only uses the best-quality electrical components on the market. Circuit breakers are made by SIEMENS and servo drives by YASKAWA. Through the selection of trusted components, DENER ensures the quality of the fiber laser machine.







AIR CONDITIONER ON THE ELECTRICAL ENCLOSURE OPTIONAL

The electrical enclosure can be equipped with an air conditioner on customer demand. This option is advised in very hot working environments.

GAS CONTROL · STANDARD

HOERBIGER GAS CONTROL

The cutting gas control system from HOERBIGER merges the features that are crucial for future-oriented laser cutting:



CUTTING GAS CONTROL

- High gas pressure
- Fast, precise gas pressure control
- High flow rate and low valve weight
- Low power consumption

These factors push the limits of speed, precision and stability in the cutting process.

GAS PRESSURE CONTROL VALVE WITH PILOT VALVES

- Maximum dynamics
- Gas-tight pilot valves
- Back-pressure resistant gas selector valves (no check valves)
- Rapid gas pressure changes
- Extremely stable gas control performance
- Entire cutting gas function in one unit
- Compact design and low weight

5.9 · SERVO MOTORS · PNEUMATIC SYSTEM · STARTING KIT

SERVO MOTORS · PNEUMATICS

SERVO MOTORS

BRAND	SUPPLY
YASKAWA	Standard
FANUC	Optional



Yaskawa

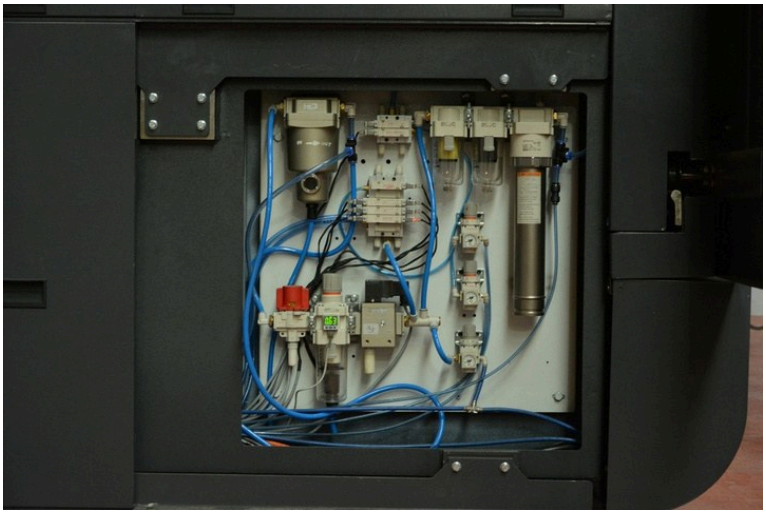


Fanuc

To improve the quality of our machines, DENER may use different brands.

PNEUMATIC SYSTEM

Through the use of FESTO pneumatic components, the management and distribution of air in the system is well done. An optional chemical dryer can also be mounted.



STARTING KIT

The fiber laser machine is delivered with a starting kit of nozzles, ceramics and protective windows.

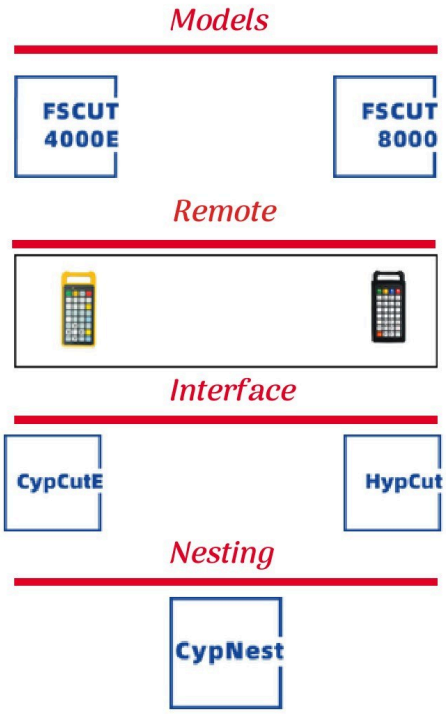


5.1.1 · CONTROL UNIT · STANDARD

CONTROL UNIT FSCUT

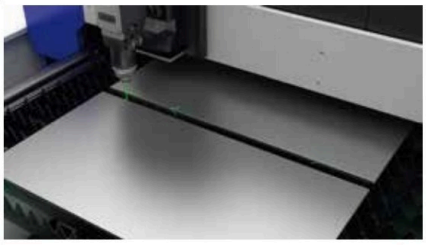
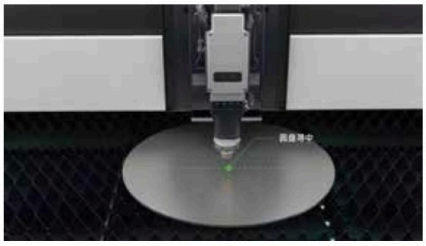
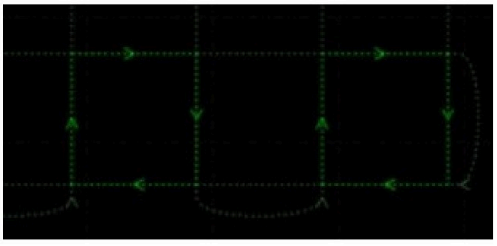
FSCUT is a cost-effective laser control system for 1.5 – 60 kW cutting. Based on EtherCAT and integrating motion control, laser control and gas control, it is widely used for sheet metal.





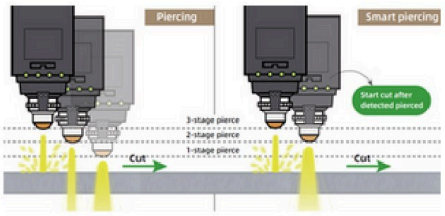
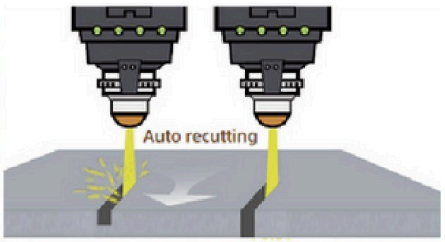
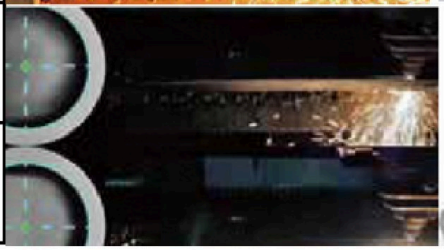
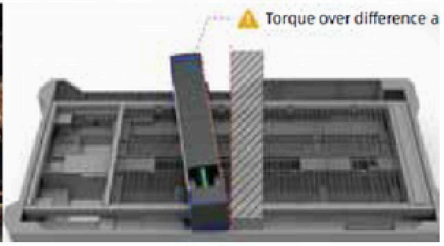
FSCUT models, remote, interface and nesting

FSCUT FEATURES

Quick CutOff		
Frame border contour		
One path flycut		
Circle center		

5.1.1 · CONTROL UNIT · STANDARD

FSCUT FEATURES

Avoid Collision	
Fast Cutting	
Auto recut (FSc8)	
Smart piercing (FSc8)	
Cutting path monitor (FSc8)	
Dual drive anti twist (FSc6)	

FSCUT 4000E AND 8000C

		FSCUT4000E	FSCUT8000C
System	Protective Visual Monitor		★
	High-Speed Find Edge	★	★
	Vertex Marking	★	★
	Circle Center	★	★
	Nest	★	CypNest
Feature	Inkjet Print	★	
	Limitless Coil	★	
	QR Code Import	★	★
	Multi-Station Machining	★	★
	Visual Remnant		★
Technique	Pierce Stage	3-Stage	9-Stage
	Fast Cutting	★	★
	Slag-Free Cutting		★
	Smooth Microjoint	★	★
	Auto Recut		★
	Smart Piercing		★

System, feature and technique functions of the FSCUT 4000E and 8000C

6.1 · SOFTWARE

CYPNEST

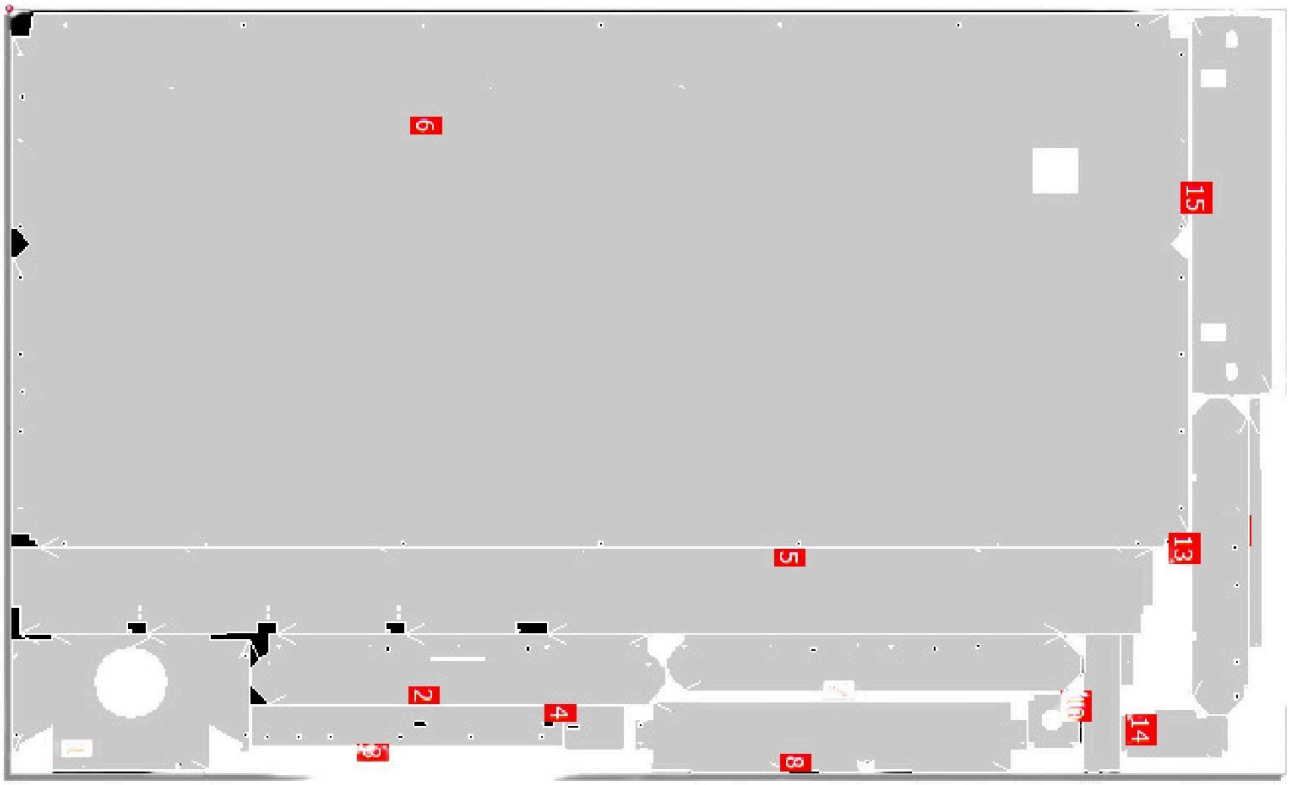
CypNest is a nesting software designed for the CypCut / HypCut sheet laser cutting system. It integrates advanced functions of drawing modification, quick nesting, toolpath generation, analysis reports and more to meet your production needs.

Drawing modification and import — intelligent parts recognition, drawing error identification and optimization. Supports DWG and DXF file formats. Supports batch creation and import of parts via an Excel table.

Nesting strategy — a powerful algorithm engine of high efficiency and production rate; no limit on sheet and part quantity in nesting; supports manual and automatic nesting; supports free-form sheet nesting.

Toolpath optimization — a smart toolpath strategy to minimize travel length; smooth operation for toolpath adjusting; multiple common-line options to meet your production needs.

Production assist — generates analysis reports of production and pricing. Technique and cutting parameters perfectly match the HypCut / CypCut laser cutting software.



A sheet nested in CypNest

5.1.2 · CONTROL UNIT · OPTION

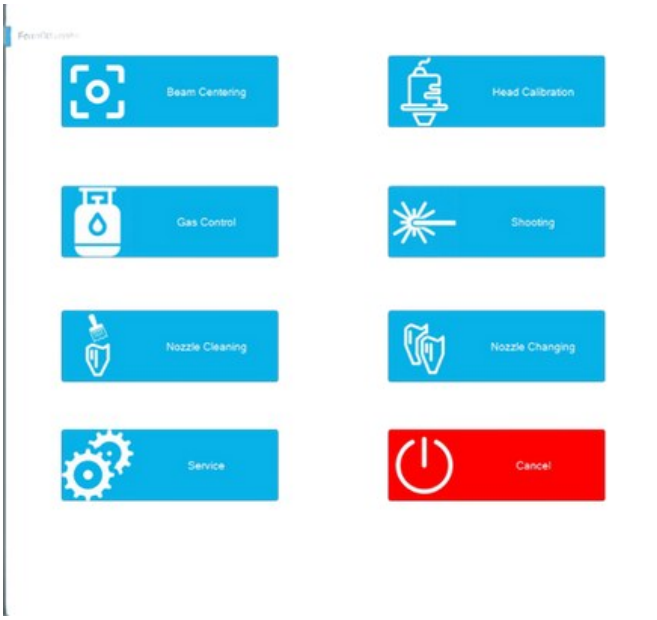
CONTROL UNIT FANUC

- High-speed Z-axis gap control
- Laser assist gas selection
- Laser assist gas pressure control
- Laser power, frequency and duty control commands
- Laser parameter overrides
- High-speed piercing
- Piercing time control
- Laser power control in real time
- Edge machining for high-quality contours
- Start-up function to stabilize cutting
- Laser restart
- Cutting parameter library
- High-speed positioning (ping-pong or frog jump)
- Fly-cut operation
- Integrated laser diagnostics
- 3D laser machining
- Tool center point programming
- Bevel cutting compensation
- W-axis gap control
- HMI customization



DENER INTERFACE AND FANUC

- Cutting technology tables
- 2D graphical view of the nesting and parts
- Alarm messages
- Live cutting area monitor
- Power control data
- Edge control data
- Sheet measurement
- Work offset
- Auto beam centering
- Full keyboard with emergency stop
- Feed rate joystick
- Industrial mouse
- Camera and monitoring



6.2 · SOFTWARE

METALIX CAD/CAM

The cncKad software provides an integrated system covering the complete cycle of CNC operations, and includes the following modules:

AutoNest — automatic true-shape nesting for optimal material utilization, an automatic DXF-to-NC solution and detailed data reports.

Post-processing — advanced post-processors generate efficient NC code, including macros, optimized tool path, minimal turret rotation and support for machine operations such as oiling, vacuum and ram rate.

Simulation — graphic simulation of the NC program enables easy editing while viewing the processed sheet. The program is automatically checked for errors such as missing parameters, clamp errors, over-travel errors and more. The NC-to-DFT option imports legacy NC programs and simulates them as well.

DNC — the DNC module supports uploading and downloading NC files to the machine, as well as batch loading and extracting NC files from the machine controller.

Import and export (optional) — 3D integration with SolidWorks®, Solid Edge®, Autodesk® Inventor® and ProE® through a real-time associative link, and reading files directly from AutoCAD®.

cncKad also offers parametric programming, advanced processing technologies and support for a wide range of machines.

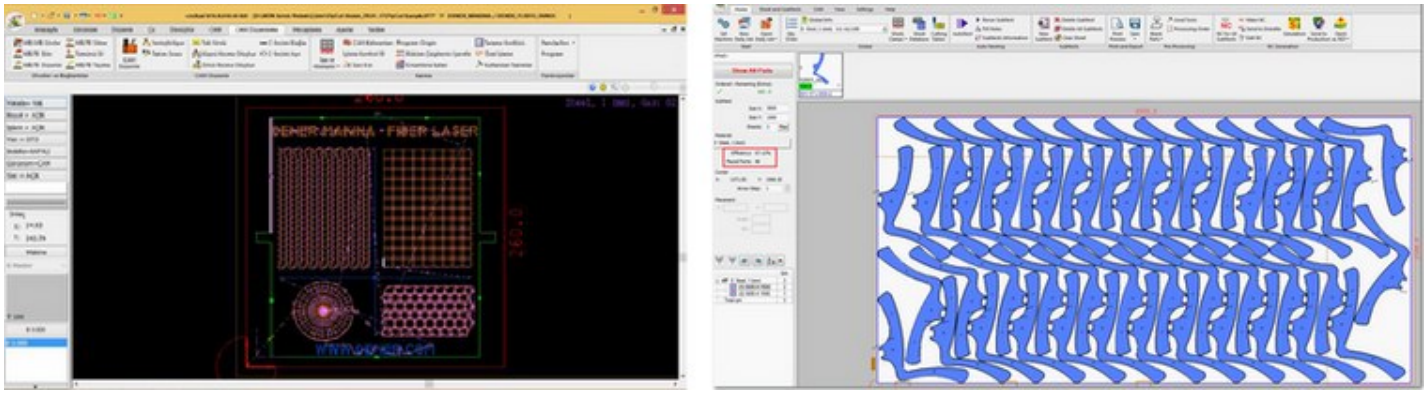


Figure 6 — cncKad

5.12 · CUTTING PARAMETERS

CUTTING PARAMETERS (CAPACITY)

Maximum cutting thickness by laser source power. Plate thicknesses are given in the nearest 1/16", with Dener's millimetre value.

Material	3,000 W	4,000 W	6,000 W	8,000 W
Mild steel	11/16" (18 mm)	13/16" (20 mm)	1" (25 mm)	1-3/16" (30 mm)
Stainless steel	5/16" (8 mm)	3/8" (10 mm)	9/16" (15 mm)	1" (25 mm)
Aluminum	1/4" (6 mm)	5/16" (8 mm)	9/16" (15 mm)	1" (25 mm)
Copper	3/16" (4 mm)	1/4" (6 mm)	5/16" (8 mm)	3/8" (10 mm)
Brass	3/16" (4 mm)	1/4" (6 mm)	3/8" (10 mm)	1/2" (12 mm)

Material	12,000 W	15,000 W	20,000 W
Mild steel	1-3/8" (35 mm)	1-9/16" (40 mm)	1-15/16" (50 mm)
Stainless steel	1-3/16" (30 mm)	1-3/8" (35 mm)	1-15/16" (50 mm)
Aluminum	1-3/16" (30 mm)	1-3/8" (35 mm)	1-9/16" (40 mm)
Copper	1/2" (12 mm)	1/2" (12 mm)	1/2" (12 mm)
Brass	9/16" (15 mm)	11/16" (18 mm)	13/16" (20 mm)

Cutting thicknesses may vary due to factors such as material quality, operator settings and cutting speed.

INCLUDED WITH EVERY NOVOTECH MACHINE

- ✓ Made in Turkey with European components
- ✓ CSA-approved electrical panel
- ✓ 1-year warranty
- ✓ **7-day money-back guarantee** (in-stock machines only)
- ✓ Spare parts & technical support readily available in Canada
- ✓ **Order Today with Just 10% Down — 0% Down Financing Available**
- ✓ **Installation and training available**

TECHNICAL DATA

OTHER TECHNICAL SPECIFICATION

NET CUTTING SIZE	1,530 × 3,050 MM 60.2 × 120.1″	2,040 × 4,050 MM 80.3 × 159.4″	2,040 × 6,050 MM 80.3 × 238.2″
X, Y transfer system		Rack & pinion	
Z transfer system		Ball screw	
Z axis stroke		13.8″ (350 mm)	
Max. speed X, Y axis		394 ft/min (120 m/min)	
Max. speed Z axis		164 ft/min (50 m/min)	
X, Y acceleration		1.5 G	
Z acceleration		1 G	
Axis positioning precision		±0.002″ (±0.05 mm)	
Axis repeatability precision		±0.0012″ (±0.03 mm)	
Shuttle change time	16 s	22 s	30 s
Cutting assistance gas		Oxygen, nitrogen, air	
Approx. weight	18,739 lb (8,500 kg)	22,487 lb (10,200 kg)	27,778 lb (12,600 kg)

GFL 3015 L · GFL 4020 L · GFL 6020 L.



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TECHNICAL DATA

ELECTRIC CONSUMPTION

Source power		3,000 W	4,000 W	6,000 W	8,000 W
Consumption	kW	35.8	39.7	46.4	51.45
Core diameter	µm	50	50	50	100

Source power		12,000 W	15,000 W	20,000 W
Consumption	kW	64	80	96
Core diameter	µm	100	100	100

Total electrical consumption of the machine by laser source power; fiber core diameter of the laser source.



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STANDARD CONFIGURATION

CONFIGURATION

Choose size	**1 · 3015**	2 · 4020	3 · 6020
Resonator power	**6 kW**	Raycus / FSCUT	
Choose configuration	**1 ✓**	2	

STANDARD

- Boji cutting head
- Machine body
- Raycus resonator
- Control unit: FSCUT series
- Software: CypCut
- Exchangeable tables
- Camera for visualization of the working area
- Air cutting system
- Cutting features
- Operating kit

**Filter is not included. UPS is not included.*



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WARRANTY

STANDARD WARRANTY CONDITIONS

1. The warranty time is 12 months for mechanical parts and moving systems. Our warranty is limited to replacing parts found to be defective; consumable items are not included.
2. We reserve the right to invoice service or visits found to be necessary due to incorrect use of the machine or lack of maintenance.
3. 12 months for the laser source, except the parts listed in the following section.
4. Cooling unit, ventilation filter, compressor and air dryer if included in the offer; the pneumatic system is 12 months.
5. Consumables of the cutting head are not covered by the warranty.

EXCLUDED FROM WARRANTY

Dener excludes the machine parts described in the following articles from the warranty scope:

- All optical parts are consumables and are not covered by the warranty.
- Consumables such as nozzles, cooling water, oils and filters are not covered by the warranty.
- Malfunctions that may occur as a result of the use of bad-quality gas, cooling water or electricity provided by the customer are not covered by the warranty.
- Malfunctions that may occur in the mirrors, the laser or the cooling unit itself due to the cooling water used in the cooling unit are not covered by the warranty.
- Failures that may occur as a result of incorrect setting of cutting parameters, incorrect light line adjustment and back reflection are not covered by the warranty. Such errors occur when untrained or incompletely trained personnel use the machine; such uses are not covered under warranty.
- Using the machine for purposes other than cutting is out of warranty.
- Malfunctions that occur as a result of incomplete maintenance or careless use are not covered by the warranty. Periodic maintenance and related procedures must be carried out by the customer.
- All work done by any company other than DENER, or not authorized by DENER, and the results of this work are not covered by the warranty.

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QUESTIONS & ANSWERS

FAQ

WHAT IS THE LIFE OF A NOZZLE?

Nozzle life is mainly related to the usage of the machine. For example, wrong nozzle centering and wrong cutting parameters will damage the component faster.

CONSUMABLES?

We recommend using original consumables. Any damage to the machine related to the use of bad-quality consumables will not be covered by the warranty.

DO I NEED A NOZZLE FOR EVERY THICKNESS?

No, nozzles are usable for a certain range of thickness (check the nozzle document).

WHAT IS THE SMALLEST HOLE THAT I CAN CUT IN A PLATE?

The smallest hole diameter in a plate is half the thickness of the plate.

HOW LOUD IS THE LASER MACHINE?

70 – 75 decibels.

IS THE MACHINE DELIVERED WITH A STARTING KIT?

Yes, the machine is delivered with a set of protective lenses, nozzles and ceramics.

HOW IS THE MAINTENANCE ON A FIBER LASER?

The machine is provided with a periodic maintenance label. A fiber laser needs less maintenance than a CO2 laser.

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7 · TRANSPORT DIMENSIONS · INSTALLATION

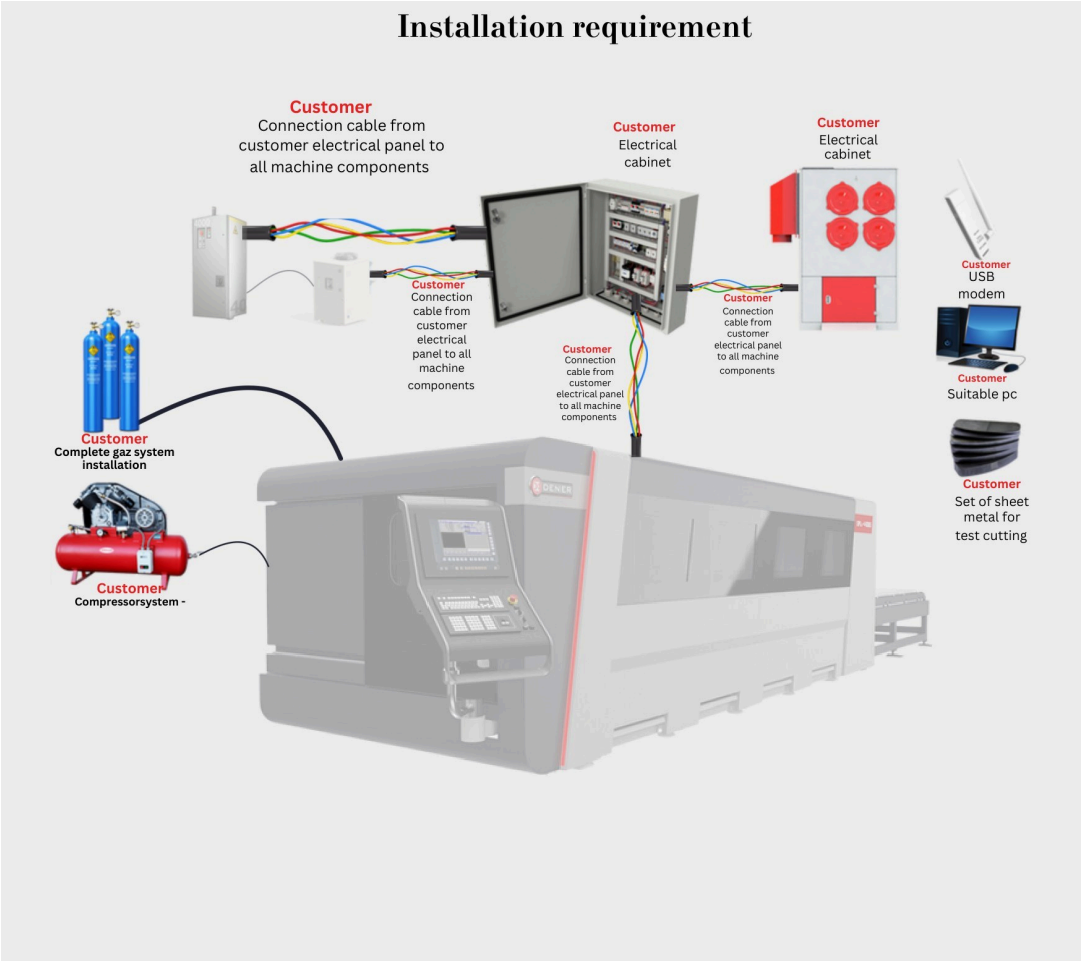
TRANSPORT

CHOOSE SIZE	TRUCKS
GFL 3015	1 × A
GFL 4020	1 × B
GFL 6020	2 × B



A — closed trailer · B — flatbed trailer

INSTALLATION REQUIREMENT



Supplied by the customer: the connection cables from the customer's electrical panel to all machine components, the electrical cabinet, the complete gas system installation, the compressor system, a suitable PC, a USB modem and a set of sheet metal for test cutting

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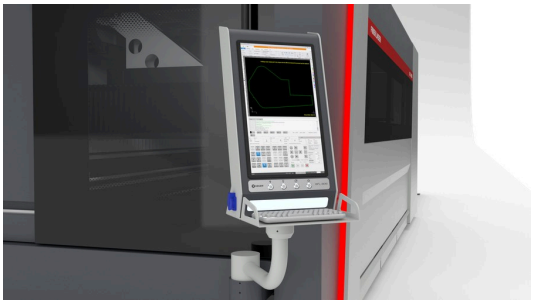
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Specifications subject to change without notice. Imperial values are converted from the manufacturer's metric data and rounded: plate thicknesses in common fractions (1/2, 1/4, 1/8, 1/16). Model numbers are manufacturer designations. Photos may show optional equipment.